

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 12: Algebraic Identities

Topic-wise short notes with solved examples

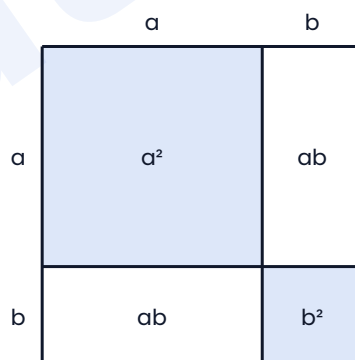
The standard identities for the square of a sum and a difference, the product of a sum and a difference, the square of a trinomial and the cubes of binomials, with applications.

1 Identity and equation

- An **identity** is an equality that is true for **all** values of its variables. $(x + 2)^2 = x^2 + 4x + 4$ is an identity.
- An **equation** is true only for certain values. $x + 2 = 5$ is true only when $x = 3$.

2 The basic identities

Identity	Expansion
$(a + b)^2$	$a^2 + 2ab + b^2$
$(a - b)^2$	$a^2 - 2ab + b^2$
$(a + b)(a - b)$	$a^2 - b^2$
$(x + a)(x + b)$	$x^2 + (a + b)x + ab$



A square of side $a + b$ has area $a^2 + 2ab + b^2$

Example 2.1 Expand $(3x + 5y)^2$ and $(4a - 3b)^2$.

SOLUTION

$9x^2 + 30xy + 25y^2$ and $16a^2 - 24ab + 9b^2$

Example 2.2 Expand $\left(2x + \frac{1}{2x}\right)^2$.

SOLUTION

$$4x^2 + 2(2x)\left(\frac{1}{2x}\right) + \frac{1}{4x^2}$$
$$4x^2 + 2 + \frac{1}{4x^2}$$

Example 2.3 Find the product $(5m + 2n)(5m - 2n)$.

SOLUTION

$$25m^2 - 4n^2$$

Example 2.4 Find the product $(x + 7)(x - 3)$.

SOLUTION

$$x^2 + (7 - 3)x + (7)(-3)$$
$$x^2 + 4x - 21$$

3 Quick calculation

- Write the number as a sum or difference involving a round number.

Example 3.1 Evaluate 103^2 , 97^2 and 105×95 .

SOLUTION

$$103^2 = 10000 + 600 + 9; 97^2 = 10000 - 600 + 9; 105 \times 95 = 100^2 - 5^2$$

10609, 9409 and 9975

Example 3.2 Evaluate $(8.3)^2 - (1.7)^2$.

SOLUTION

$$(8.3 + 1.7)(8.3 - 1.7) = 10 \times 6.6$$

66

Example 3.3 Evaluate 52×53 .

SOLUTION

$$(50 + 2)(50 + 3) = 2500 + 250 + 6$$

2756

4 Relations between the identities

- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $a^2 + b^2 = (a + b)^2 - 2ab = (a - b)^2 + 2ab$
- $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = \left(x - \frac{1}{x}\right)^2 + 2$

Example 4.1 If $a + b = 8$ and $ab = 15$, find $a^2 + b^2$.

SOLUTION

$$64 - 30$$

34

Example 4.2 If $a - b = 5$ and $ab = 14$, find $a^2 + b^2$ and $(a + b)^2$.

SOLUTION

$$a^2 + b^2 = 25 + 28 = 53; (a + b)^2 = 25 + 56$$

53 and 81

Example 4.3 If $x + \frac{1}{x} = 6$, find $x^2 + \frac{1}{x^2}$.

SOLUTION

$$36 - 2$$

34

Example 4.4 If $x - \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$.

SOLUTION

$$9 + 2 = 11; \text{ then } 11^2 - 2$$

11 and 119

5 Square of a trinomial

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

- For a negative term, carry its sign into the products.

Example 5.1 Expand $(x + 2y + 3z)^2$.

SOLUTION

$$x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6zx$$

Example 5.2 Expand $(2a - b + c)^2$.

SOLUTION

$$4a^2 + b^2 + c^2 - 4ab - 2bc + 4ca$$

Example 5.3 If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$.

SOLUTION

$$81 - 52$$

$$29$$

6 Cubes of binomials

- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$
- $x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

Example 6.1 Expand $(2x + 3)^3$.

SOLUTION

$$8x^3 + 3(4x^2)(3) + 3(2x)(9) + 27$$

$$8x^3 + 36x^2 + 54x + 27$$

Example 6.2 Expand $(a - 2b)^3$.

SOLUTION

$$a^3 - 6a^2b + 12ab^2 - 8b^3$$

Example 6.3 If $a + b = 4$ and $ab = 3$, find $a^3 + b^3$.

SOLUTION

$$a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 64 - 36$$

$$28$$

Example 6.4 If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.

SOLUTION

$$27 - 9$$

$$18$$

Example 6.5 Evaluate $(99)^3$.

SOLUTION

$$(100 - 1)^3 = 1000000 - 30000 + 300 - 1$$

970299

7 Common mistakes

- Writing $(a + b)^2 = a^2 + b^2$ and leaving out $2ab$.
- Writing $(a - b)^2 = a^2 - b^2$.
- Taking the last term of $(a - b)^2$ as $-b^2$. It is $+b^2$.
- Forgetting to square the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.
- Using $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 + 2$. The sign is minus.
- Missing the factor 3 in the middle terms of a cube.